Response to the editor and reviewers

Manuscript ARRAY-D-26-03529
Mathias Valla
12 October 2026
Original title: Sparse farther-sighted trees can improve forest accuracy–time tradeoffs
Revised title: A few farther-sighted trees can improve shallow forests

Dear Dr Hazarika,

Thank you for the opportunity to revise the manuscript. The article has been substantially improved in both evidence and presentation. It now follows a clearer question: can a few farther-sighted members improve a shallow forest without requiring every member to search farther? The results show positive mean gains from sparse replacements under both feature policies, examples of small mixtures outperforming much larger greedy forests with less measured fitting work, and a higher benchmark-mean accuracy for the expanded family when both competitors are tuned. The abstract, introduction, discussion, and conclusion now develop these findings directly; execution detail has been reduced and the main limitations are brought together in Section 6.1.

Reviewer 1's concerns have shaped the revised evaluation: repeated paired partitions, deeper and unrestricted capacities, matched-depth unpruned CART and farther-sighted controls, training-only tuning of both conventional and mixed-capable families, and paired statistical analysis. Implementation and timing issues have also been corrected. The contribution is an empirical assessment of selective use of established bounded split search, rather than a claim to a new lookahead algorithm. The detailed response below retains the methodological safeguards and numerical qualifications behind the more direct presentation.

Following your suggestion, I would welcome your guidance on an editor-approved transfer, possibly to *Pattern Recognition Letters*, whose concise format suits this study. An earlier version was declined there; I would therefore seek the receiving editor's agreement to reconsider this substantially revised manuscript. I would also consider other suitable journals offering publication without a mandatory article processing charge (APC), including a subscription route, with the applicable terms confirmed before I consent to transfer.

Reviewer 2's report appears to concern a different manuscript: it describes a numerical-aware LogBERT extension for telecommunications monitoring, whereas this submission concerns decision-tree induction and bootstrap ensembles. I respectfully ask you to verify whether that report was attached to the correct submission. Below I identify the mismatch point by point, without attributing its cause or attempting to answer unrelated technical questions.

Response to Reviewer 1

R1.1. Depth-three restriction and practical scalability

Comment: The depth limit is unrealistic for practical random forests, and lookahead is computationally prohibitive at realistic depths.

I agree that a depth-three comparison alone cannot establish practical competitiveness. The new protocol retains that controlled setting but additionally tests maximum final depth six and no depth limit on the same datasets and partitions. CART and one-, two-, and three-sighted trees have the same common hyperparameters and allowed total depth, with no post-pruning in either implementation. The fixed-architecture forest comparison also matches feature subsampling, bootstrap multiplicities, and ensemble size. A separate training-only selection experiment is

applied to both conventional and mixed-capable forest families, rather than tuning only the baseline. Its shared structural grid includes deeper configurations; sharing that grid does not imply equal candidate counts or computation. Both stages are complete and their full analysis has passed validation.

The Introduction and the new related-work subsection "Tree-size regularization" clarify why a shallow regime is nevertheless scientifically relevant. Zhou and Mentch (2023), *Trees, forests, chickens, and eggs: when and why to prune trees in a random forest*, provide direct evidence for tree-size regularization, particularly at low signal-to-noise ratios (doi:10.1002/sam.11594). Duroux and Scornet (2018), *Impact of subsampling and tree depth on random forests*, analyze early termination and subsampling in quantile-split regression forests and examine tree-size choices empirically (doi:10.1051/ps/2018008). Their small-tree empirical alternative also changes resampling. These are conditional results, with regression, leaf-count, and resampling settings that differ from ours. They motivate studying shallow members but do not validate depth three as a general optimum for Gini classification forests. The cited consistency theory also concerns growing tree sizes, not a fixed eight-leaf limit.

Our depth-three design limits each member to at most eight leaves; it does not limit the aggregated forest to eight prediction regions or make a large ensemble as interpretable as one small tree. The revised manuscript acknowledges that deeper forests can reduce underfitting and improve prediction. Shallow-tree results from boosting are not used to justify bootstrap-forest performance. Thus this literature clarifies the research scope, while the added deep and tuned controls address the reviewer's empirical objection.

Section 3 now separates the local optimization horizon k from the final tree-depth limit D. A two-generation search is restored at each actual node and can build a deeper tree; its horizon does not grow with D. I corrected the previous complexity discussion: complete split-assignment counts are not the work performed by the additive recursion. A uniform candidate-count abstraction gives $E_h \leq m+2mE_{h-1}$, before sorting and sample-partition costs. This still produces expensive search, but does not prove that every modern non-greedy method is unscalable.

The revised claims remain limited to the fixed PMLB sample. Unlimited final depth on small tasks is not evidence of scalability to large/high-dimensional modern workloads. Three-sighted members are now included in the completed repeated fixed and tuned experiments, rather than represented only by historical single-split screening. Relevant locations: Sections 3.1–3.3, 4.2, the depth results in Section 5 and Figure 1, and the limitations in Section 6.1.

R1.2. Single split, small differences, and lack of statistical analysis

Comment: Without cross-validation, repeated runs, tuning, or significance tests, the observed gains cannot be separated from random variation.

I agree that the original screening needed a stronger evaluation. The revised protocol uses five paired stratified outer partitions per dataset, with changing estimator/bootstrap seeds and an outer test fraction of one quarter. Fixed, unpruned CART and sighted models are compared with matching common settings. Separately, conventional RF, the one-/two-sighted family, and the inclusive one-/two-/three-sighted family all undergo inner cross-validation using the same outer-training rows and fold indices. All share depths $\{3, 6, \infty\}$, leaf minima $\{1, 5\}$, both feature policies, and ensemble sizes $\{20, 40, 60, 100\}$; both expanded families include the entire RF candidate space. One configuration per family is chosen using training data alone, then separately refitted before outer-test evaluation. The fixed comparisons isolate replacements; the tuned comparisons evaluate tuned families on both sides. Larger candidate spaces and their full selection work are reported separately from the cost of the selected refit.

Here RF denotes the conventional greedy family: the tuned feature policy permits either feature-subsampled random forests or all-feature bagged CART. Inner cross-validation is stratified when every observed training class has at least two examples. A singleton or single-class training partition uses a documented ordinary shuffled three-fold fallback; this is not reliable rare-class validation. The 200-member fixed curves are descriptive and are not tuning options for any family.

The balanced-accuracy sensitivity uses the library's mean recall over classes present in the scoring partition. A predicted class absent from that partition has undefined recall and is excluded from this mean; the library warns about this situation. This is not a fit failure, and the metric cannot establish recall for absent rare classes. The primary accuracy criterion and the common split protocol are unchanged.

The completed analysis first averages paired repeat effects within each dataset. It reports equal-dataset mean and median differences, wins/ties/losses, balanced-accuracy sensitivity, unadjusted percentile dataset-bootstrap intervals, and two-sided Wilcoxon tests with one Holm adjustment across eleven prespecified fixed and tuned contrasts. All other grid points are descriptive. Repeated overlapping holdouts are not counted as independent datasets. The bootstrap estimates a mean effect and the signed-rank test a location hypothesis, with its usual symmetry qualification; those are distinguished. Known related benchmark families and named constructed/synthetic tasks receive separate sensitivity analyses. A nonsignificant result is not interpreted as equivalence.

The fixed and training-selected stages are now complete on all 57 datasets and five paired outer partitions, with no missing or skipped task. The fixed stage comprises 1,710 structural blocks, 5,130 single-tree scores, and 136,800 forest scores. The selected stage provides three separately refitted models for each of 285 outer tasks, adding 855 scores. Independent full analysis reconstructed stored fixed and inner-validation forest predictions and constituent fitting costs and independently reconstructed the validation winners. Single-tree and direct-refit scores received range and checkpoint consistency checks, not independent prediction reconstruction. The complete analysis now applies the single prespecified eleven-contrast Holm adjustment. Figures 2–3 and Tables 1–2 use these completed records, rather than the preceding tuned-versus-untuned evaluation.

The depth controls show where the benefit is strongest. For all-feature single trees, horizons two/three have descriptive mean differences from CART of 2.77/3.54 points at depth three, 0.39/0.59 at depth six, and $-5.13/-8.29$ without a depth limit. At 100 members, pure all-feature horizon-two/three forests have mean differences of 1.90/2.34 points at depth three but $-6.84/-9.68$ without a depth limit. The square-root feature setting differs: its unrestricted pure-forest mean differences are 1.08/0.78 points. We therefore do not generalize the all-feature depth pattern to conventional feature-subsampled forests or claim that shallow members are universally preferable. Selected RF chooses final depth six or unrestricted depth in 219 of the 285 outer dataset–repetition selections. The conditional tree-size literature motivates the shallow question, not depth three as a general optimum, and boosting results are not treated as bootstrap-forest evidence.

For the primary depth-three all-feature comparison, replacing ten of 100 CART members with horizon-two/three trees gives mean gains of 1.06 [0.45, 1.77] and 1.27 [0.61, 2.02] points. Both medians are zero, with 27/21/9 and 28/14/15 wins/ties/losses, and joint $p_H = 0.00996/0.02066$. With square-root features, gains are 0.42/1.04 points, with intervals [0.07, 0.78] and [0.46, 1.74], and $p_H = 0.09539/0.00150$. Thus three of the four primary shallow contrasts retain $p_H < 0.05$. At depth three with all features, mean member-fitting work rises from 0.0677 s for the matched greedy forest to 0.2293/8.0933 s, or 3.39/119.52 times the control; the balanced-accuracy gains are 1.15/1.33 points. Under equal weighting of 48 known task families, the accuracy gains are

0.77/0.91 points with intervals [0.25, 1.45] and [0.32, 1.63]. On the 38 tasks outside the named synthetic/constructed set, they are 0.53/0.63 points with intervals [0.06, 1.22] and [0.07, 1.35].

Without a depth limit, the four primary replacement mean differences are $0.05/-0.02$ points with all features and 0.22/0.28 with square-root features. Their medians are zero and no corresponding adjusted signed-rank value is below 0.05; this does not establish equivalence. The poorer all-feature unrestricted pure-tree and pure-forest results, despite greater realized tree depth, are consistent with overfitting. Section 6 presents this as an explanation to investigate, not a tested causal mechanism.

With training-only tuning on both sides, mean accuracies are 78.99% for RF, 79.96% for the one-/two-sighted family, and 80.56% for the inclusive family. The inclusive family's mean accuracy gain is 1.5693 points [0.2991, 3.2767], with median zero, 28/11/18 wins/ties/losses, and joint $p_H = 0.48838$. The one-/two-sighted family's gain is 0.9674 points [−0.0668, 2.2979], with median 0.00617 points, 29/8/20 wins/ties/losses, and $p_H = 1$. Inclusive versus one-/two-sighted gives 0.60 points [−0.11, 1.63], 18/23/16 wins/ties/losses, and $p_H = 1$. The inclusive family's balanced-accuracy gain over RF is 1.60 points. The positive inclusive mean interval is not contradicted by its non-rejected signed-rank location null; these summaries have different estimands. We do not describe the result as no improvement or equivalence. However, the inclusive mean gain becomes 1.03 points [−0.27, 2.90] under the 48-family weighting and 0.53 [−0.91, 2.77] on the 38 tasks outside the named synthetic/constructed set. These intervals cross zero and limit a broad generalization claim. The corresponding one-/two-sighted gains are 0.61 points [−0.32, 2.02] and 0.38 [−0.66, 2.09]. Moreover, 133 of 285 inclusive selections are pure forests and 152 are mixed (146 two-horizon and six three-horizon mixtures). The selected-family effect cannot be attributed solely to sparse mixtures.

All families receive the same common structural and ensemble-size grid, not equal computation: their candidate spaces contain 48, 288, and 768 configurations. Standalone cross-validation fitting work averages 2.271, 112.079, and 2,671.049 s for RF, one-/two-sighted, and inclusive families, including banks not represented in the eventual selected model. Direct refit wall times are 0.0626, 0.9518, and 12.9374 s, including construction/bootstrap preparation but excluding prediction. The expanded families' refit ratios to RF are 15.20 and 206.61; these are not selection-work ratios. The sums of measured workflow stage wall times are 3.405, 118.035, and 2,693.803 s. These exclude checkpoint I/O, startup, downloads, and idle gaps and are not uninterrupted end-to-end measurements. Physical shared-bank accounting counts common work once, rather than summing the standalone family charges. The larger search costs remain an important practical qualification to the positive tuned mean effects.

The text directly after Figure 2 now explains the positive descriptive result. At depth three with all features, a 20-member forest containing 15 CART and five two-sighted trees achieves 74.063578% mean accuracy at 0.095460218 s of mean member-fitting work, compared with 73.020360% at 0.13382638 s for 200 greedy members. This is 1.043218 percentage points higher accuracy with 28.6686% less measured fitting work. Even the 20-member mixture with 18 CART and two two-sighted trees reaches 73.3727% at 0.047004 s, exceeding that larger greedy forest's mean accuracy at lower work. These observed points show the value of allocating computation to composition as well as ensemble size.

The scope of this comparison is precise: scores and member-fit durations are averaged over repetitions within each dataset and then equally across datasets. Fitting work excludes construction, bootstrap preparation, prediction, and model selection; it is not an end-to-end wall-time measure. The pure two-sighted 20-member forest remains the highest observed mean-accuracy point below the 0.7 s reference (74.7904% at 0.3381338 s). The mixtures therefore improve on the specified larger greedy comparator, not on every pure forest. Neither the reference line nor interpolation proves exact equal-time superiority or validates a per-dataset budget policy. Section

6.1 retains these limits, the benchmark sensitivities, and the implementation/hardware timing scope. The conclusion emphasizes the demonstrated higher benchmark-mean accuracy at lower member-fitting work, without restoring the ad hoc utility or attributing the gains to a proven diversity mechanism.

Relevant locations: Sections 4.1–4.4 and 5, Tables 1–2, Figures 1–3, the complete CSV results, paired-comparison table, dataset manifest, and tuning records. The original single-split gains are not reused as uncertainty evidence.

R1.3. Limited novelty and no new search efficiency

Comment: Lookahead induction is old, and the manuscript does not overcome its computational bottleneck.

I agree and have changed the title, abstract, introduction, related work, and conclusion accordingly. Section 2 now treats Norton's overlapping search and Murthy and Salzberg's two-level LOOK rule as direct prior art. It also acknowledges Esmeir and Markovitch's bagged non-greedy learners and computation-per-tree versus committee-size question, discusses stepwise lookahead forests, and positions the study against modern cached/dynamic-programming and hybrid methods, including DL8.5, MurTree, STreeD, and SPLIT. The recent literature on objective and tuning choices is also discussed.

The contribution is the empirical finding that selective farther sight can improve shallow ensembles: sparse replacements raise mean accuracy, and small mixtures can outperform much larger greedy forests with less member-fitting work. The separate tuned comparison shows the value of allowing farther-sighted candidates when both families receive training-only selection. Implementation detail has been reduced in the article; it does not alter the established bounded optimization criterion or its worst-case horizon dependence. There is no claim to a new search algorithm or asymptotic improvement. A solver benchmark against the modern methods has not been performed, and Section 6.1 identifies that comparison as future work.

R1.4. Ad hoc utility heuristic

Comment: The accuracy–time utility is simplistic and ad hoc.

I agree. I removed the logarithmic utility and its break-even coefficients. The revised study reports predictive differences and measured fitting costs directly, including the observed higher-accuracy, lower-work example in Figure 2 described under R1.2. Section 6.1 explains the remaining resource-policy question: a per-dataset budget experiment would require training-only selection, accounting for selection cost, and a check of final fitting feasibility. The descriptive curve is not presented as that experiment.

Additional corrections identified during revision

The revision also corrects three issues not explicitly raised in the report:

1. Historical mixtures joined prefixes of separately fitted forests and could reuse bootstrap positions. New mixtures replace distinct positions, with exactly one member at each position and correct class-probability alignment.

2. Historical mixed times were allocated from pure-bank averages. New controlled-ensemble costs sum the actual timings of the included members; selected-model whole-refit wall times

are retained separately. Timing boundaries, excluded preparation/prediction, baseline selection cost, and concurrent workload are stated.

3. The preceding custom greedy learner differed from standard CART in zero-gain stopping and bootstrap sample-count semantics. The new protocol uses actual library CART as its one-sighted baseline, allows feasible zero-gain splits in sighted trees, and uses identical weighted bootstrap draws. A verification harness establishes RF-bank prediction agreement within absolute tolerance 10^{-12}, with zero relative tolerance, at every considered ensemble size, and tests depth-matched interaction recovery without relying on a zero-gain stopping difference. This check validates the greedy-bank comparator; it does not establish bit-identical numerical and tie-breaking behavior for the farther-sighted learners.

The corrected protocol is described in the manuscript; implementation checks and historical details remain in the reproducibility documentation. Historical aggregates without retained repeat-level evidence are separated from the new complete evaluation and do not support its inference.

Response to Reviewer 2

Thank you for the time devoted to reviewing. The supplied report appears to describe a different manuscript from ARRAY-D-26-03529. I respectfully ask the editor to check its assignment and would welcome any corrected report intended for this submission. The following points explain the apparent mismatch without assuming its cause:

R2.1. *Comment: A numerical-aware extension of LogBERT for telecom monitoring; positioning against BARTPredict and MTRC.*

This manuscript does not propose or evaluate LogBERT, a telecommunications system, or a proactive LLM framework. It studies Gini-based decision-tree split optimization and bootstrap forests. I request confirmation of the report's assignment.

R2.2. *Comment: Congestion ground truth, chronological splitting, normalization, threshold calibration, and numerical embedding are insufficiently specified.*

There is no congestion ground truth, numerical embedding, or anomaly threshold in the submitted study. Its labels are PMLB classification targets. The revised paper specifies stratified outer partitions and training-only inner model selection; a chronological telecom protocol would address a different research problem.

R2.3. *Comment: Check the anomaly-score formulation.*

No anomaly score is defined. The manuscript's criterion is weighted terminal Gini impurity, now specified by an additive recursion in Section 3.1.

R2.4. *Comment: Insufficient uncertainty, statistical analysis, per-class results, and evidence that the model predicts.*

Although the report appears to concern another model, uncertainty and statistical assessment are relevant general concerns and are addressed by the repeated, paired analysis described under R1.2. The revised study reports held-out classification accuracy and balanced accuracy. It makes no telecommunications forecasting or class-specific deployment claim; a telecom per-class validation cannot be supplied for this study.

R2.5. *Comment: Reduce or support claims about cyberattack detection, generalization, real-time performance, and production readiness.*

The manuscript makes no cyberattack, real-time monitoring, or production-readiness claim. Its generalization statements are confined to held-out performance on the stated benchmark sample. I would be grateful for any corrected report intended for this submission and would address it in a further response.

Thank you for considering the revision and for clarifying the apparent report mismatch.

Yours sincerely,
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